## Introduction

The dataset used for this visualisation project is the Global Historical Climatology Network–Monthly (GHCN-M) dataset [1], chosen as it is one of the most widely used, rigorously curated, and continuously updated climate archives available. Maintained by the National Oceanic and Atmospheric Administration (NOAA), the GHCN-M provides monthly temperature records for thousands of meteorological stations worldwide, many of which contain more than a century of observations. Its consistent formatting and structured layout make it well suited for a large-scale visualisation.

This project focuses specifically on regional temperature trends in Europe over the period 1924–2024. Europe has a dense and geographically diverse network of meteorological stations, allowing for comparative analysis across regions while retaining statistical consistency. The objective is to transform the raw GHCN-M data into a file with an analysis-ready format and then use said file to build an interactive dashboard that reveals both long-term annual temperature evolution and evolving monthly patterns.

The dashboard would be designed not only to display climate data but also to support exploration of spatial and temporal variation across Europe. A map-based interface would allow users to select individual stations, while supplementary time-series plots would reveal both long-term temperature trends and local anomalies. Through this combination of spatial selection and temporal visualisation, users can investigate how temperature patterns have evolved at both regional and station-specific scales.

## Method

The GHCN-M dataset is distributed in two fixed-width text formats: an inventory file (*ghcnm.tavg.v*.inv*) containing station metadata and a data file (*ghcnm.tavg.v*.dat*) containing monthly temperature observations. Because these files cannot be read directly into analytical software without explicit column definitions, a custom Python script, ghcnm.py, was written to parse, clean, and restructure both sources into a single, analysis-ready comma-separated variable (.csv) file.

The inventory file was parsed using fixed column boundaries taken directly from NOAA's documentation [2], which specify the location of the station identifier, latitude, longitude, elevation, and station name. These specifications were passed to *pandas.read_fwf*, producing a *DataFrame* in which station identifiers were stored as strings to preserve their formatting. Parsing the temperature file required a more detailed approach. Each record contains the station identifier, year, an element code such as *TAVG*, and twelve groups of monthly values (*v*), each accompanied by data-measurement (*dm*), quality-control (*qc*), and data-source (*ds*) flags. To accurately decode this structure, the script programmatically generated all column specifications for the twelve months, resulting in fields {*v1, dm1, qc1, ds1*} through to {*v12, dm12, qc12, ds12*}. To maintain focus on monthly mean temperature rather than other climate variables, only rows associated with the *TAVG* element were retained after loading the file with *read_fwf*.

The GHCN-M dataset stores monthly temperature values as integer hundredths of degrees Celsius, with -9999 denoting missing data. A conversion function was applied across all value fields to convert valid integers into degrees Celsius and replace missing values with NaN. This produced a clean numerical dataset suitable for aggregation, computation, and visual analysis. Following the conversion, spatial and temporal filters were applied. The inventory table was restricted to stations located within Europe by selecting latitudes between 35° and 71° north and longitudes between –25° and 41° east. These geographic bounds were derived based on approximations made in the online geospatial tool, BoundingBox [3]. The temperature table was limited to the years 1924–2024 to form a consistent 100-year analysis window. These filtered tables were then merged on the station identifier, yielding a unified dataset containing metadata, monthly temperatures, and derived metrics. An annual mean temperature was calculated for each station-year combination by averaging all 12 monthly values. The resulting dataset was exported as *ghcnm_europe_1924_2024.csv*, which served as the input for the visualisation dashboard.

The dashboard was developed using the Dash framework in Python, with Plotly providing the interactive graphical components. A second script, *app.py*, loaded the cleaned dataset and prepared the spatial component of the visualisation by extracting one unique coordinate entry per station. A MapLibre-based scatter map was generated using Plotly's map functions, displaying station locations with red markers and enabling automatic clustering in densely sampled regions to mitigate visual clutter. Each marker encoded its corresponding station identifier in the *customdata* field, allowing user interactions on the map to cause updates elsewhere in the interface. Two callback functions were implemented: one to generate an annual time-series plot for the selected station and another to construct a monthly climatology plot by reshaping the dataset into long form. Both visualisations updated dynamically in response to user selections on the map, providing an interactive, easy-to-use environment for exploring spatial and temporal temperature patterns across Europe.

## **Results**

The final visualisation presents a clean and responsive interface that allows users to explore a century of temperature variability across Europe. The map-based overview successfully displays the spatial distribution of stations while ensuring visual clarity through automatic clustering, which prevents marker overlap in densely monitored regions such as central Europe. Clicking any station reveals two complementary plots: an annual temperature trend and a monthly climatology profile. Together, these views highlight both long-term warming patterns and seasonal variation.

The annual temperature plot shows clear upward trends for many stations, especially those in central and northern Europe, where the rate of warming has been maintained over the past several decades. The monthly climatology plots reveal distinct seasonal patterns that remain broadly consistent across stations, though some locations show stronger winter warming or more pronounced summer peaks than others. The ability to toggle dynamically between different stations makes it easy to compare patterns across the continent, demonstrating how local climate responds differently to broader global changes. The interactive dashboard thus effectively distils a large, complex dataset into an accessible form that promotes exploration.

## **Discussion**

The main motivation for developing a visualisation of the GHCN-M dataset is the scale and structural complexity of the data itself. With thousands of stations, each recording observations for over a century and each month encoded with multiple flags, the dataset is too complex to interpret through basic tables or static summaries alone. The archaic fixed-width formatting adds another layer of complexity, requiring careful parsing before even basic trends can be identified. Visualisation is thus the most effective means for revealing spatial variation, long-term temporal patterns, and seasonal dynamics, all of which are crucial in understanding the dataset but remain inaccessible to non-specialists without a graphical representation.

Python, pandas, and Dash were chosen because of their flexibility, reproducibility, and robust support for large datasets. Pandas provides reliable tools for fixed-width parsing and data cleaning, which were essential for reconstructing monthly temperature values and aligning them with station metadata. Several alternative visualisation tools were tested during development, but Dash and Plotly proved to be the easiest to learn given the project's limited timeframe. Their ability to create interactive browser-based visualisations without the need for additional software made them ideal for the task. The built-in support for linking map selections to dynamically updating plots also aligned well with the project's focus on exploratory analysis.

The resulting visualisation can be used to perform a variety of analyses. Users can track long-term temperature trends at individual stations, compare seasonal patterns across regions, and investigate how spatial variation relates to geographic features like longitude, latitude, and elevation. Interactive callbacks make it possible to move fluidly between spatial exploration and temporal analysis, allowing for a deeper engagement with the data than static charts would allow.

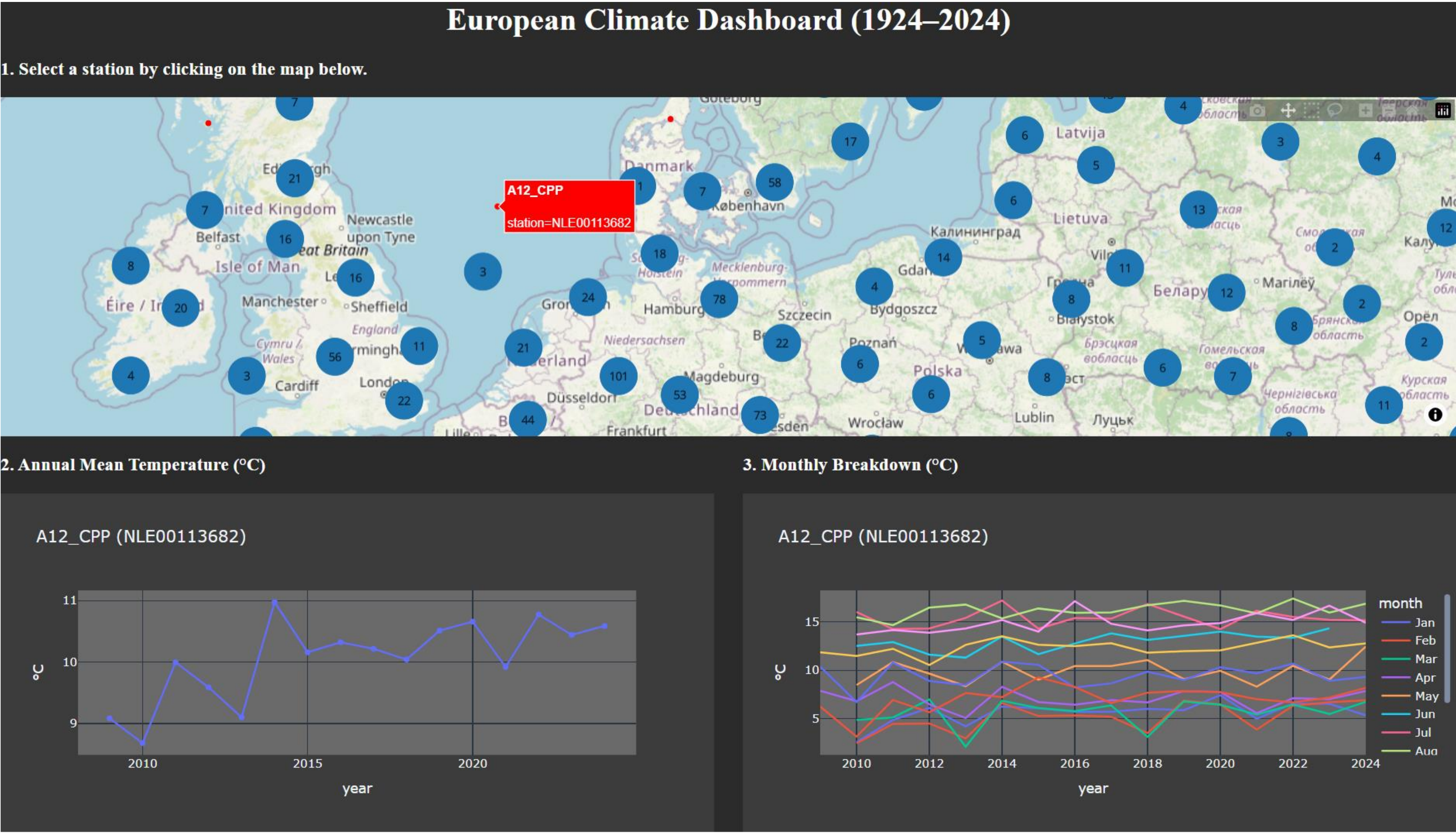
The novelty of this implementation lies in the integration of large-scale climate data processing with a station-level interactive interface. While many climate dashboards exist, they often focus on station-aggregated national or global datasets. In contrast, this visualisation retains the full granularity of individual station records, allowing users to explore climate behaviour at the level where the data is originally measured. The underlying parsing workflow is technically complex, requiring precise handling of fixed-width data, monthly subfields, various flag types, and integer-encoded temperatures. This technical complexity justifies a custom implementation over pre-existing tools.

The choice of encoding channels followed established conventions in climatology while prioritising perceptual clarity for the reader. Spatial position acts as the primary encoding on the map, with coloured markers indicating station locations (red) and station clusters (blue) used to reduce visual clutter. Temporal trends are expressed through line charts, which naturally communicate continuity and long-term change. The monthly climatology view highlights seasonal structure and variability with coloured lines, while the pop-up windows from hover interactions reveal more precise values. Together, these encodings provide a broad overview and a detailed examination of climate patterns.

## **Conclusion**

There are several strengths to this visualisation. Its interactive design makes it easy to navigate a large and complex dataset without overwhelming the user, while the coordinated spatial and temporal views create a coherent narrative about climate change across Europe over the years. The preprocessing pipeline of ghcnm.py ensures that the dataset is accurate and consistent, improving the reliability of the visual outputs. However, there are still some limitations to the visualisation. The development timeframe limited the scope of variables included, meaning the dashboard focuses only on mean temperature (TAVG), excluding other indicators such as maximum and minimum temperatures or precipitation. Some stations have gaps in their records, most likely due to operational interruptions hindering the recording of observations, resulting in discontinuities in the time series. Furthermore, the map could be improved upon by adding geographic context such as country-level temperature and/or altitude fields, using shading.

Despite these limitations, the visualisation succeeds in transforming a dense and technically challenging dataset into an accessible analytical tool. It demonstrates how targeted data cleaning, careful choice of encodings, and interactive design can reveal structure and trends that would otherwise be inaccessible.


European Climate Dashboard (1924–2024)
1. Select a station by clicking on the map below.
A12_CPP
station=NLE00113682
2. Annual Mean Temperature (°C)
3. Monthly Breakdown (°C)
A12_CPP (NLE00113682)
°C
year
month
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug

**References**


[1]. NOAA National Centers for Environmental Information, "Global Historical Climatology Network Monthly (GHCN-M), Version 4", *Noaa.gov*, 2024. https://www.ncei.noaa.gov/pub/data/ghcn/v4/ (accessed Nov. 24, 2025).

[2]. Documentation, "Global Historical Climatology Network Monthly (GHCN-M), Version 4," *Noaa.gov*, 2019. https://www.ncei.noaa.gov/pub/data/ghcn/v4/readme.txt (accessed Nov. 27, 2025).

[3]. Klokan Technologies GmbH, "*Bounding Box Tool", 2025*. https://boundingbox.klokantech.com/ (accessed Dec. 1, 2025).


**Operation Instructions**

1. *Download Required Files & Place in The Same Working Directory:*

   - ghcnm.py (data-processing script)
   - app.py (visualisation application)
   - Raw GHCN-M files:
     - ghcnm.tavg.v4.0.1.20251123.qcf.inv (station inventory)
     - ghcnm.tavg.v4.0.1.20251123.qcf.dat (temperature data)
   - libs.txt (required python libraries)

2. *Install Required Python Libraries via Command prompt*:

```
pip install -r libs.txt
```

3. *Run the Data-Processing Script to Generate the Cleaned Dataset:*

```
python ghcnm.py
```

(Outputs *ghcnm_europe_1924_2024.csv in working directory.)*

4. *Launch the Dashboard by Running:*

```
python app.py
```

5. *Open the Local Session Initiated by Dash in a Browser:*

"`Dash is running on http://127.0.0.1:8050/`"

6. *Using the Visualisation:*

   - Zoom in on any blue cluster and click a red station marker to load its data.
   - The annual average and monthly temperature plots update automatically after selection.
   - Hover over markers in any plot to view additional details.
   - The monthly climatology plot supports interactive legend filtering: single-click a month label to hide its line; double-click to isolate that month.